



APRIL/2025

REGIONAL MARITIME UNIVERSITY
ICT DEPARTMENT

BME28B/BMT28B/BNS28B/BEE28B/BCE28B/BCS28B/BIT28B

END OF SEMESTER EXAMINATION

BCME 101 - COMPUTER STUDIES 1

INSTRUCTIONS:

ANSWER ALL QUESTIONS IN SECTION A AND ONLY TWO QUESTIONS IN SECTION B

TIME ALLOWED: 2 HOURS 30 MINUTES

INDEX NUMBER:

QUESTION 1

SECTION A

Multiple Choice Questions 1 to 60

[1 mark each]

1. UNIVAC is

- A. Universal Automatic Computer
- B. Universal Array Computer
- C. Unique Automatic Computer
- D. Unvalued Automatic Computer

2. The basic operations performed by a computer are

- A. Arithmetic operation
- B. Logical operation
- C. Storage and relative
- D. All the above

3. Microprocessors as switching devices are for which generation computers

- A. First Generation
- B. Second Generation
- C. Third Generation
- D. Fourth Generation

4. What is the main difference between a mainframe and a super computer?

- A. Super computer is much larger than mainframe computers
- B. Super computers are much smaller than mainframe computers
- C. Supercomputers are focused to execute few programs as fast as possible while mainframe uses its power to execute as many programs concurrently
- D. Supercomputers are focused to execute as many programs as possible while mainframe uses its power to execute few programs as fast as possible.

5. ASCII and EBCDIC are the popular character coding systems. What does EBCDIC stand for?

- A. Extended Binary Coded Decimal Interchange Code
- B. Extended Bit Code Decimal Interchange Code
- C. Extended Bit Case Decimal Interchange Code
- D. Extended Binary Case Decimal Interchange Code

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6. The brain of any computer system is
- ALU
 - Memory
 - CPU
 - Control unit
7. The two kinds of main memory are:
- Primary and secondary
 - Random and sequential
 - ROM and RAM
 - All of above
8. Integrated Circuits (ICs) are related to which generation of computers?
- First generation
 - Second generation
 - Third generation
 - Fourth generation
9. Chief component of first generation computer was
- Transistors
 - Vacuum Tubes and Valves
 - Integrated Circuits
 - None of above
10. The main electronic component used in first generation computers was
- Transistors
 - Vacuum Tubes and Valves
 - Integrated Circuits
 - None of above
11. A dumb terminal has
- an embedded microprocessor
 - extensive memory
 - independent processing capability
 - a keyboard and screen
12. In most IBM PCs, the CPU, the device drives, memory expansion slots and active components are mounted on a single board. What is the name of this board?
- Motherboard
 - Breadboard
 - Daughter board
 - Grandmother board
13. A modern electronic computer is a machine that is meant for
- Doing quick mathematical calculations
 - Input, storage, manipulation and outputting of data
 - Electronic data processing
 - Performing repetitive tasks accurately
14. Software in computer
- Enhances the capabilities of the hardware machine
 - Increase the speed of central processing unit
 - Both of above
 - None of above
15. What is the responsibility of the logical unit in the CPU of a computer?
- To produce result
 - To compare numbers
 - To control flow of information
 - To do math's works
16. Abacus was the first
- electronic computer
 - mechanical computer
 - electronic calculator
 - mechanical calculator
17. People often call _____ as the brain of computer system
- Control Unit
 - Arithmetic Logic Unit
 - Central Processing Unit
 - Storage Unit

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18. Which is used for manufacturing chips?
- Bus
 - Control unit
 - Semiconductors
 - A and B only
19. Which generation of computer is still under development
- Fourth Generation
 - Fifth Generation
 - Sixth Generation
 - Seventh Generation
20. From which generation operating systems were developed?
- First
 - Second
 - Third
 - Fourth
21. The first firm to mass-market a microcomputer as a personal computer was
- IBM
 - Super UNIVAC
 - Radio Shaks
 - Data General Corporation
22. Properly arranged data is called
- Field
 - Words
 - Information
 - File
23. A computer consists of
- A central processing unit
 - A memory
 - Input and output unit
 - All of the above
24. Multi user systems provided cost savings for small business because they use a single processing unit to link several
- Personal computers
 - Workstations
 - Dumb terminals
 - Mainframes
25. What are the three decisions making operations performed by the ALU of a computer?
- Greater than
 - Less than
 - Equal to
 - All of the above
26. The word processing task associated with changing the appearance of a document is
- Editing
 - Writing
 - Formatting
 - All of above
27. Which statement is valid about computer program?
- It is understood by a computer
 - It is understood by programmer
 - It is understood user
 - Both a & b
28. The difference between memory and storage is that memory is _____ and storage is _____
- Temporary, permanent
 - Permanent, temporary
 - Slow, fast
 - All of above
29. A compiler is a translating program which
- Translates instruction of a high level language into machine language
 - Translates entire source program into machine language program
 - It is not involved in program's execution
 - All of above

30. What is required when more than one person uses a central computer at the same time?

- A. Light pen
- B. Mouse
- C. Digitizer
- D. Terminal

31. The term gigabyte refers to

- A. 1024 bytes
- B. 1024 kilobytes
- C. 1024 megabytes
- D. 1024 gigabyte

32. Machine language is

- A. Machine dependent
- B. Difficult to program
- C. Error prone
- D. All of above

33. A byte consists of

- A. One bit
- B. Four bits
- C. Eight bits
- D. Sixteen bits

34. Modern Computers are very reliable but they are not

- A. Fast
- B. Powerful
- C. Infallible
- D. Cheap

35. Programs designed to perform specific tasks is known as

- A. system software
- B. application software
- C. utility programs
- D. operating system

36. Computer operators

- A. writes computer programs for specific problems
- B. operate the device which input and output data from the computer
- C. normally require a college degree in computer science
- D. all of the above

37. What is a compiler?

- A. A compiler does a conversion line by line as the program is run
- B. A compiler converts the whole of a higher level program code into machine code in one step
- C. A compiler is a general purpose language providing very efficient execution
- D. None of the above

38. _____ computers are also called personal computers

- A. Mainframe Computer
- B. Mini Computers
- C. Micro Computers
- D. Super Computers

39. Which of the following is not input unit device?

- A. scanner
- B. camera
- C. plotter
- D. digitizer

40. Which of the following require large computers memory?

- A. Imaging
- B. Graphics
- C. Voice
- D. All of Above

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41. Which of the following is machine independence program?
- High level language
 - Low level language
 - Assembly language
 - Machine language
42. Which one is the largest space?
- kilobyte
 - petabyte
 - terabyte
 - gigabyte
43. The brain of any computer system is
- Control Unit
 - Arithmetic Logic Unit
 - Central Processing Unit
 - Storage Unit
44. The BIOS is the abbreviation of _____.
- Basic Input Output System
 - Best Input Output System
 - Basic Input Output Symbol
 - Base Input Output System
45. Most of the first generation computers were
- Special purpose computers
 - General purpose computers
 - Both of above
 - None of above
46. Operating system, editors, and debuggers comes under?
- System Software
 - Application Software
 - Utilities
 - None of the above
47. One computer that is not considered a portable is
- Minicomputer
 - Laptop computer
 - Notebook computer
 - All of above
48. Which of the following is required when more than one person uses a central computer at the same time?
- Terminal
 - Light pen
 - Digitizer
 - Mouse
49. Which of the following is not an output device?
- Scanner
 - Printer
 - Flat Screen
 - Touch Screen
50. This computer component holds processed information before it is output.
- RAM
 - ROM
 - bus
 - ports
51. Converting $(153)_{10}$ to base 8 yields which of the following results?
- 107
 - 132
 - 701
 - 231
 - 153
52. Converting $(1010111)_2$ to base 8 yields which of the following results?
- 531
 - 721
 - 44D
 - 135
 - 127
53. 10111 is the two's complement representation of
- 23
 - 9
 - 7
 - +22e. +7

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54. An intranet is a public network and an extranet is a private network.
A. True
B. False
55. Secondary storage is another name for memory
A. True
B. False
56. These computers are arranged from least powerful to most powerful: microcomputer, midrange computer, mainframe, and supercomputer.
A. True
B. False
57. The shared resources that the operating system manages include the CPU, memory, storage device, and the printer.
A. True
B. False
58. An input device that reads printed text and graphics and transfers them to a computer in digital form.
A. digital camera
B. scanner
C. MICR
D. RFID readers
59. A device that enables a computer to communicate over telephone lines.
A. Repeater
B. Modem
C. Switch
D. Wireless access point
60. This coding system is designed to support international languages like Chinese and Japanese.
A. ASCII
B. Unicode
C. EBCDIC
D. ANSI

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SECTION B

ANSWER ONLY TWO (2) QUESTIONS IN THIS SECTION

QUESTION 2

Write a **pseudocode** that ask the user to enter a positive integer, obtain the number from the user, and determines whether the given number is even or odd. [20 marks]

QUESTION 3

- a) State **five** input devices. [5 marks]
- b) State **five** output devices. [5 marks]
- c) Explain how computer data travels over telephone lines. [4 marks]
- d) List the three services the internet provides its users. [6 marks]

QUESTION 4

Write a flow chart to convert temperate in Fahrenheit to Celsius using formula below: [20 marks]

$$(32^{\circ}\text{F} - 32) \times 5/9 = 0^{\circ}\text{C}$$

QUESTION 5

- a) What is the difference between data and information? [4 marks]
- b) State three uses of a Super Computer [6 marks]
- c) To secure files on your PC, so they are unreadable to a hacker, who might gain access to your PC, what type of encryption (public or private key) would be most appropriate? Explain. [6 marks]
- d) List two precautions that individuals can take when purchasing items via an online auction to avoid loss due to online auction fraud. [4 marks]

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